



OPEN An insulator target detection algorithm based on improved YOLOv5

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Drone inspections are widely utilized in the detection of insulators in power lines. To address issues with traditional object detection algorithms, such as large parameter counts, low detection accuracy, and high miss rates, this paper proposes an insulator detection algorithm based on an improved YOLOv5 model. Firstly, in the backbone and neck networks, a lightweight CSP-SCConv module is employed to replace the original CSP-Darknet53 module, thereby reducing the parameter count and enhancing the feature extraction capabilities. Secondly, to broaden the image receptive field and improve feature fusion, a Receptive Field Block (RFB) model is incorporated into the neck network, replacing the original Spatial Pyramid Pooling Fast (SPPF) module. Additionally, a Lattice Structured Kernel (LSKBlock) attention mechanism is appended at the end of the neck network to further obtain richer semantic information. Finally, to flexibly improve the accuracy of bounding boxes of different sizes and enhance the robustness of the model, an $\alpha CIOU$ loss function is utilized to replace the original Complete Intersection Over Union (CIOU) loss function. Experimental results demonstrate that the improved YOLOv5 model achieves a mean Average Precision (mAP) precision of 95.60%, with a parameter count of 18.36 M and a computational load of 30.10G, respectively. The Precision (P) and Recall (R) are 88.10% and 95.20%, providing strong support for deployment on mobile devices for real-time detection.

Keywords YOLOv5, Insulator, CSP-SCConv, RFB, LSKBlock

With the vigorous development of the power industry, the scale of high-voltage transmission lines is continually expanding. The stable and safe operation of these power lines provides a robust guarantee for uninterrupted power supply¹. As a critical component of power lines, insulators serve to support the lines and prevent current leakage to the ground. However, insulators can be impacted by various external factors, such as strong winds, hail, freezing rain, and other uncontrollable forces, leading to insulator damage and affecting the stability of the power grid². Consequently, regular inspections of insulators are essential to ensure the reliability of power supply. Traditional methods for insulator detection rely heavily on manual patrols, which are both time-consuming and costly. Under adverse weather conditions and complex backgrounds, the effectiveness of manual patrols diminishes further, potentially endangering the personal safety of patrol personnel³. With the ongoing advancements in deep learning, object detection algorithms have become the predominant approach. Deploying these algorithms on drone equipment for real-time detection can significantly enhance inspection efficiency.

Currently, there are two categories of object detection algorithms based on deep learning: single-stage object detection algorithms, such as SSD⁴, YOLOv1⁵, YOLOv2⁶, YOLOv3⁷, and the YOLOv4⁸ series; and two-stage object detection algorithms, such as R-CNN⁹, Fast R-CNN¹⁰, and Faster R-CNN¹¹. For the task of insulator target detection, many researchers have proposed numerous detection algorithms. Liu et al.¹² uses a single-stage SSD object detection algorithm, adopting aerial photographs of insulators and washers as the training dataset. It utilizes MnasNet to replace the original feature extraction network and employs multi-feature map fusion using two scales to increase the image's receptive field. Although the precision of this method improves, its parameter size is relatively large. Song et al.¹³ proposes a CSH-YOLOv5 object detection algorithm, replacing the original CSP-Darknet53 module with a CSHNet module in the backbone network to improve model detection accuracy. Additionally, it optimizes the backbone and neck networks of YOLOv5, but the detection effect under complex backgrounds is poor. Chen et al.¹⁴ introduces a CA attention mechanism module into the backbone network to enhance feature extraction capabilities and uses an Enhanced Intersection Over Union (EIOU) loss function

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to improve the accuracy of insulator recognition. This method increases the mAP precision value, accuracy, and recall rate, but the model structure is redundant. Ni et al.¹⁵ proposes an improved Faster R-CNN insulator detection algorithm, preprocessing the collected insulator dataset and introducing the Concept-ResNet-V2 network into the backbone network to optimize the network model structure. However, the model is large and has certain deficiencies in terms of computation. Liu et al.¹⁶ proposes an improved YOLOv5 drone object detection algorithm, first employing the C3Ghost network as the backbone to reduce network parameters. Secondly, it uses a small object decoupling detection head in the detection head to improve detection accuracy, and finally, uses EIOU to optimize the image aspect ratio to speed up model convergence, but the model has a higher rate of missed detections under complex backgrounds. Lu et al.¹⁷ uses the GhostNetv2 network as the backbone of the model to reduce the computational complexity of the model; meanwhile, it adds an adaptive weight reassignment module in the feature fusion network to reduce the loss of feature information. However, this method has lower detection accuracy when detecting large objects. Hu et al.¹⁸ introduces BiFPN (Bidirectional Feature Pyramid Network) into the feature fusion network, thereby improving the positioning ability of insulator detection targets. At the same time, it adds the SPD (Spatial-to-Depth Convolution) module to the feature fusion network, which reduces the loss of key feature information. Finally, it introduces the SimAM attention mechanism at the end of the backbone network to strengthen the model's focus on feature information, yet the parameter size and computational load of this model are high, making it unsuitable for deployment on mobile edge terminals.

Through the analysis of the above research methods, for insulator target detection where the aspect ratio is large and orientation is random, the existing method has a high rate of missed detections and poor detection accuracy. Considering the issues of large parameter count, high computational load, and model redundancy, it does not meet the requirements for deploying the algorithm on mobile devices such as drones. To solve these problems, this paper proposes an insulator target detection method based on an improved YOLOv5. The main contributions of this paper are as follows:

- (1) To reduce the number of parameters and enhance feature extraction capabilities, we have adopted a lightweight CSP-SCConv module in both the backbone and neck networks. This module not only decreases the model's parameter count but also significantly boosts detection accuracy.
- (2) To improve the image receptive field and feature fusion capabilities, we have integrated an RFB module into the neck network and added an LSKBlock attention mechanism at the end of the neck network. These enhancements enable the model to acquire richer semantic information.
- (3) To adaptively improve the precision of bounding boxes for different sizes and enhance the robustness of the model, we have utilized an improved loss function $\alpha CIOU$. Experimental results demonstrate that the proposed YOLOv5 model excels across various metrics and exhibits significant advantages over other existing models.

The remaining structure of this paper is as follows: In the next section, we introduce the basic model of the YOLOv5 algorithm and detail the improvements made in the proposed object detection algorithm. The subsequent section describes the experimental environment and data augmentation methods, providing a detailed analysis and discussion through comparisons of ablation experiments and various algorithms. Finally, the last section summarizes the content of the paper and discusses future research directions.

The improved network structure is shown in Fig. 1.

Improved YOLOv5 algorithm YOLOv5 algorithm model

YOLOv5²¹ is primarily composed of three parts: the backbone network, the neck network, and the prediction network. The backbone network consists of a series of Conv convolution blocks, CSP modules, and an SPPF feature pyramid module. Its main function is to extract feature information, converting image features into multi-level feature maps, which are then further transmitted to the neck network for feature fusion²². The neck network is comprised of Feature Pyramid Network (FPN) and Path Aggregation Network (PAN) structures; the FPN structure performs a top-down downsampling process, while the PAN structure carries out a bottom-up upsampling process. This structure facilitates the fusion of feature maps at different levels, further enhancing the accuracy of object detection. The prediction network plays a primary role before the entire model training process. First, it generates object prediction bounding boxes through the K-means clustering method and retains the bounding box with the highest confidence using the non-maximum suppression method. Next, it uses regression to determine the position and size of the object. Finally, it classifies each bounding box using an FN fully connected layer and a softmax activation function to determine whether the object exists, while also calculating the loss value of each target box using the CIOU loss function.

Improvement of the lightweight CSP-SCConv module

This article makes improvements based on the YOLOv5 network structure. The original YOLOv5 network model uses the CSP-Darknet53 structure, but due to the large parameter count of this structure, it leads to increased computational load, making it unsuitable for deployment on mobile devices such as drones. Therefore, the backbone network adopts a lightweight Cross Stage Partial - Spatial Channel Convolution (CSP-SCConv) module to replace the original CSP-Darknet53 structure, aiming to reduce the model's parameter count and enhance feature extraction capabilities. The CSP-SCConv module is designed based on improvements to the spatial and channel convolution module SCConv²³. The CSP-SCConv module mainly consists of two residual blocks and two core units, namely the Spatial Refinement Unit (SRU) and the Channel Refinement Unit (CRU). The workflow of this module is as follows: Firstly, through the output features X from the previous layer, two 1×1 convolutional processes yield feature X_0 , which is used to extract shallow feature information; subsequently, the SRU unit processes X_0 to obtain spatially refined feature X_1 ; then, the CRU unit processes X_1 to derive channel-

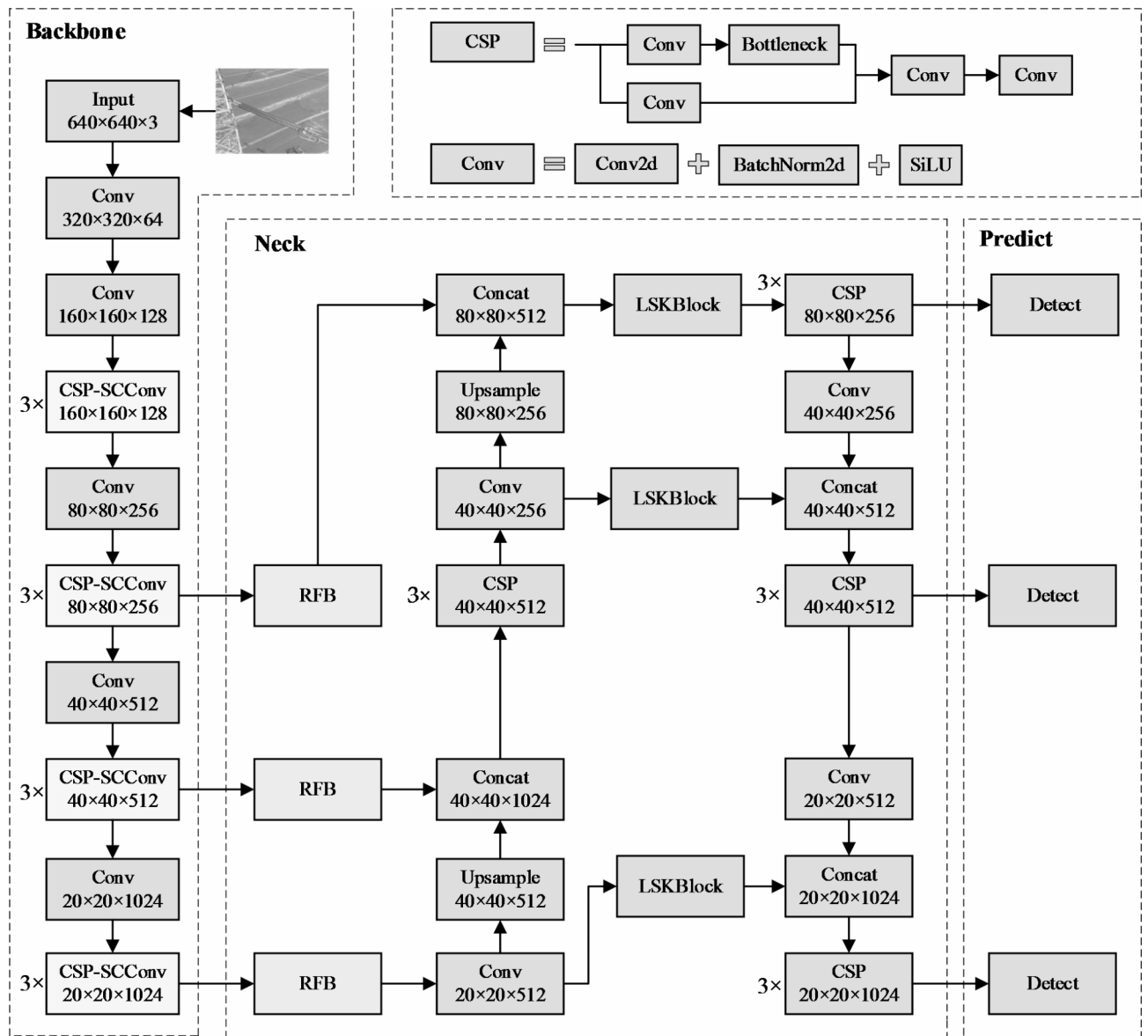
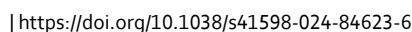


Fig. 1. Improved YOLOv5 network model.

refined feature X_2 , and convolution operations are performed to restore the number of channels; thereafter, the features after the entire SCConv module operation are concatenated with the features after the first convolution to form a residual connection block. Finally, the features obtained from an additional convolution of the input features X are stacked with the features processed by the SCConv module and undergo another convolution operation to produce the output feature Y . The specific structure of the CSP-SCConv module is shown in Fig. 2.

The structure of the Spatial Reconfiguration Unit (SRU) is illustrated in Fig. 3. Firstly, the input feature X_0 undergoes a Group Normalization (GN) operation to obtain $GN(X_0)$, which separates the feature information and simultaneously calculates the relevant weights W after normalization. Multiplying $GN(X_0)$ by W yields richer feature content and training weights. Secondly, through the Sigmoid function, the weighted feature values are constrained within the range (0, 1), and thresholds are set to further control the output. When the threshold is set to 1, information weight W_1 is obtained, and when the threshold is set to 0, non-information weight W_2 is obtained. Then, W_1 and W_2 are respectively multiplied by the input feature X_0 to produce features with high information content X_{W_1} and low information content X_{W_2} . Finally, the features X_{W_1} and X_{W_2} are reconfigured to reduce spatial redundancy, generating four new sets of features. These four sets of features are pairwise added together to produce two new sets of features X^1 and X^2 , which are then stacked to ultimately obtain the output feature X_1 .

The specific computation formulas for the Spatial Reconfiguration Unit (SRU) are shown in Eqs. (1) and (2). Here, X_{0out} is the output feature after GN processing, $X_0 \in R^{C \times H \times W}$, C is the channel axis, H and W are the height and width axes, respectively; μ and σ are the mean and variance values within the GN layer, which are approximately zero; ε is a small constant used to stabilize the output result; γ and β are the scaling factor and bias



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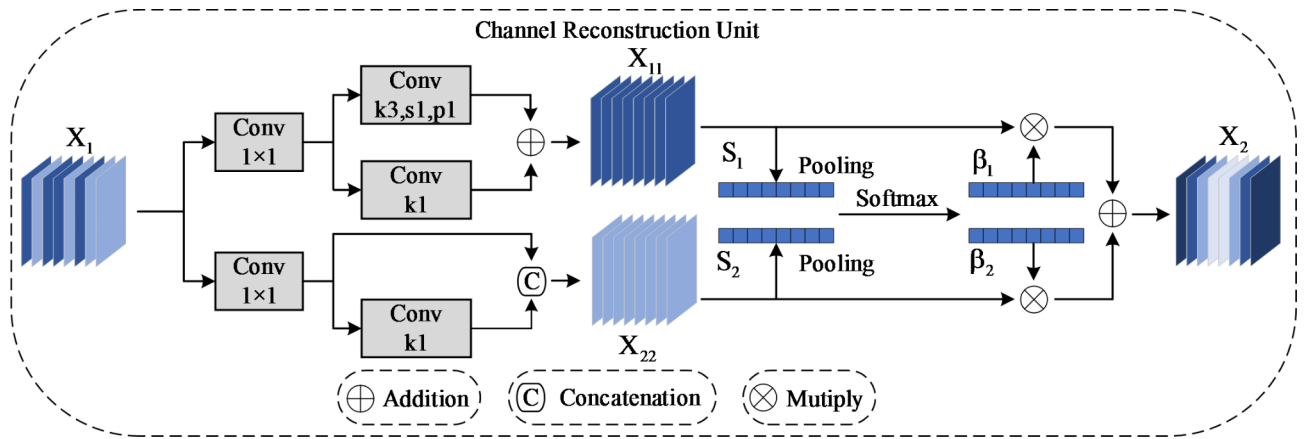


Fig. 4. Module structure of channel reconstruction unit.

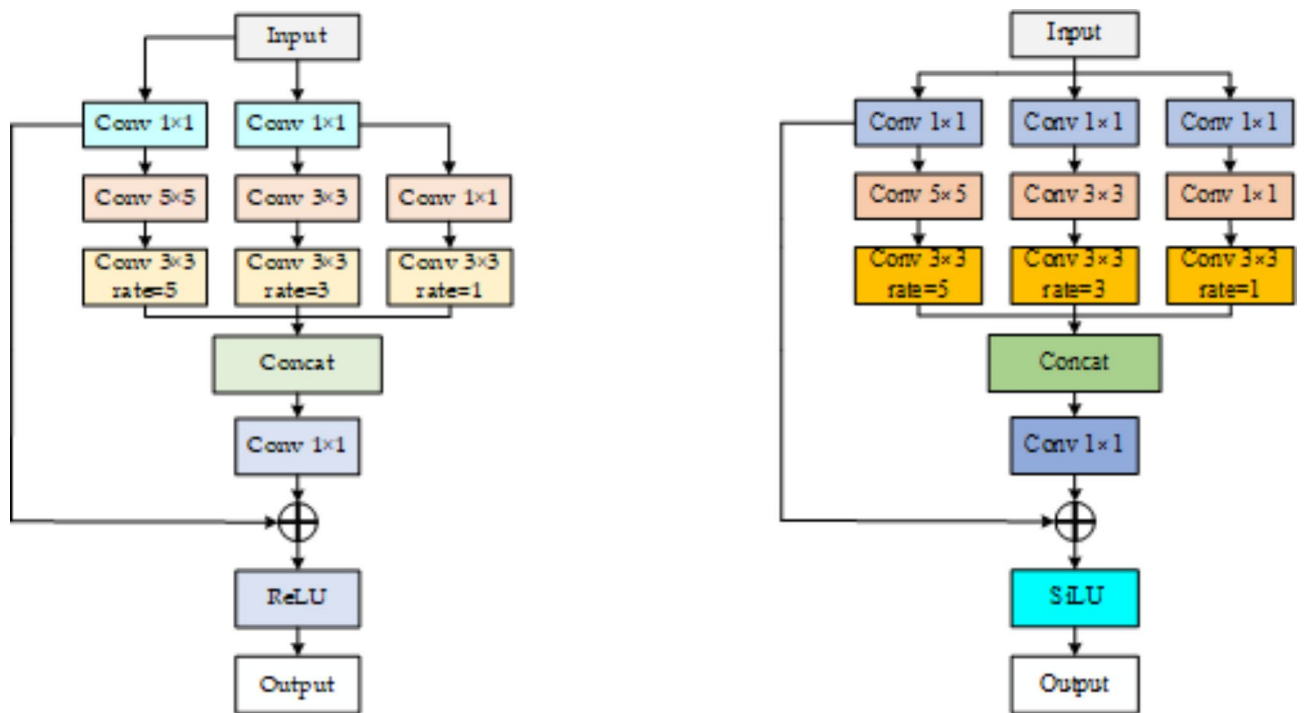


Fig. 5. (5.1) Original RFB module, (5.2) Improved RFB module.

$$S_2 = \text{Pooling}(X_{22}) = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W X_c(i, j) \quad (4)$$

$$\beta_1 = \frac{e^{s_1}}{e^{s_1} + e^{s_2}}, \beta_2 = \frac{e^{s_2}}{e^{s_1} + e^{s_2}}, \beta_1 + \beta_2 = 1 \quad (5)$$

Feature fusion network

The original YOLOv5 network model employs the SPPF feature pyramid module²⁴ at the beginning of the neck network, but this module suffers from issues such as excessive parameter count and fixed pooling kernel sizes and usage frequency. Additionally, the spatial information can be further lost and the receptive field of the image can become smaller as the image features pass through deeper convolutional layers. The RFB module is a technique designed to enhance the receptive field of convolutional neural networks. It achieves this by using dilated convolutions of different sizes to expand the receptive field of each convolutional kernel without adding extra parameters or computational costs. However, the original RFB module suffers from issues such as non-uniform receptive fields and gradient vanishing problems. The structure of the original RFB module is illustrated in Fig. 5.1. To address these issues, this paper introduces an improved RFB channel convolution module as the

starting component of the neck network. This improvement aims to enhance the receptive field and feature fusion capabilities of the image. The structure of the improved RFB module is shown in Fig. 5.2. The improved RFB module incorporates the following optimizations based on the original RFB module: An additional 1×1 convolution is added to the first layer of feature input to ensure that the model's receptive field is uniformly propagated. The original ReLU activation function is replaced with the SiLU activation function, which helps to alleviate the vanishing gradient problem in deep networks, making the model easier to train.

The design of the RFB module aims to effectively enhance the network's receptive field through two layers of convolution kernels of different sizes and one layer of dilated convolution kernels with varying dilation rates. Specifically, the module first uses three 1×1 convolution kernels to perform initial feature extraction on the input features. Subsequently, it employs 1×1 , 3×3 , and 5×5 convolution kernels to further integrate the features from the previous layer. Furthermore, the RFB module utilizes three convolution kernels with different dilation rates to simulate the changes in receptive field size and eccentricity found in the human visual system. This design allows for a more diverse and refined receptive field while keeping the total number of model parameters unchanged, thereby significantly improving the model's recognition accuracy. Finally, the module concatenates the three output features from the previous layer and processes them through a 1×1 convolution. It also connects the input end with the result processed by a 1×1 convolution to form a residual network structure, effectively preventing the problem of gradient explosion. Additionally, it adopts the SiLU activation function to replace the traditional ReLU activation function to process the output features from the previous layer, thereby further enhancing the training efficiency of the model.

To further enhance the model's ability to capture key features in images and extract rich contextual feature information, this paper embeds three LSKBlock attention mechanism modules in the Neck network. The LSKBlock attention mechanism adaptively adjusts the receptive field according to different targets, effectively covering a larger range of input information, thus significantly improving the network's perception of different targets. The specific structure is shown in Fig. 6. The module first generates two different feature maps V1 and V2 using a 5×5 convolution kernel and a 7×7 convolution kernel with a dilation rate of 3, respectively. Secondly, 1×1 convolution kernels are used to process V1 and V2, adjusting their channel numbers to make them consistent and stacking them together. Then, average pooling, max pooling, 7×7 convolution, and Sigmoid activation functions are sequentially applied to generate the corresponding weights $\hat{S}$. Finally, the weights $\hat{S}$ are multiplied with the feature maps V1 and V2, stacked together, and then multiplied with the initial input feature X to obtain the final output Y.

Improvement of the loss function

The YOLOv5 network model employs CIOU as the bounding box regression loss function. CIOU takes into account the overlap area, distance between the center points, and diagonal distance of the bounding boxes, allowing it to more accurately measure the similarity between two bounding boxes, especially in cases where the shapes and sizes of the targets are irregular²⁵. The CIOU loss function is calculated using the following formula:

$$v = \frac{4}{\pi^2} \left(\arctan \frac{w^{gt}}{h^{gt}} - \arctan \frac{w}{h} \right)^2 \quad (6)$$

$$\beta = \frac{v}{(1 - IOU) + v} \quad (7)$$

$$L_{CIOU} = 1 - IOU + \frac{\rho(b, b^{gt})}{c^2} + \beta v \quad (8)$$

Here, v is the correction factor used to measure the similarity of aspect ratios; w^{gt} and h^{gt} are the width and height of the ground truth bounding box, respectively; w and h are the width and height of the predicted bounding box, respectively; β is the weighting coefficient; IOU represents the ratio of the intersection to the union of the ground truth and predicted bounding boxes; $\rho(b, b^{gt})$ represents the Euclidean distance between the centers of the ground truth and predicted bounding boxes; c represents the diagonal distance of the minimum enclosing

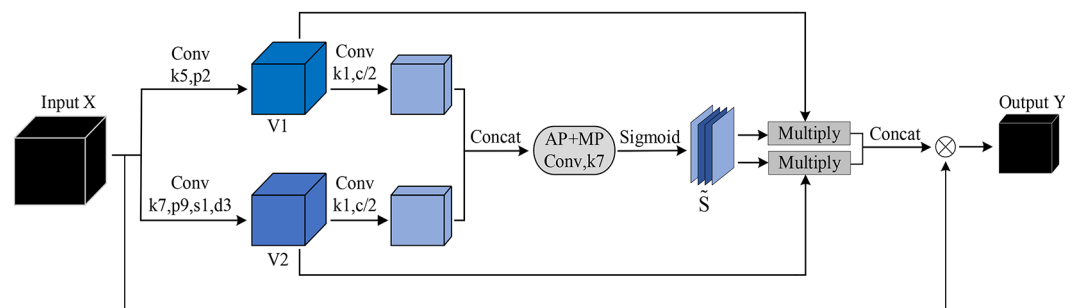


Fig. 6. LSKBlock Attention mechanism module.

box of the ground truth and predicted bounding boxes. The parameters used in the CIOU loss function are shown in Fig. 7.

The CIOU loss function is not sufficiently sensitive when dealing with small objects and objects with imbalanced aspect ratios. When computing the loss function for small objects, the distance difference between the ground truth bounding box and the detected bounding box is small, making it difficult to effectively distinguish between the two boxes. When detecting objects with imbalanced aspect ratios, the CIOU loss function may lead to unstable gradient updates, affecting the convergence and performance of the model^{26,27}. To address these issues, this paper introduces a new parameter α based on the CIOU loss function. By adjusting the value of the parameter α , the precision of bounding boxes of different sizes and the robustness of the model can be enhanced. Firstly, using the Box-Cox transformation, the ordinary IOU loss is extended to αIOU loss, with the calculation formula shown in Eq. (9). When α approaches zero, $\lim_{\alpha \rightarrow 0} L_{\alpha IOU} = -\log(IOU)$. In the cases where $\alpha \rightarrow 0$ and $\alpha > 0$. Then, by introducing a power penalty term in the formula, the αIOU is further generalized to a general form, as shown in Eq. (11). Finally, based on the derivation of the αIOU formula, it is further extended to an $\alpha CIOU$ loss function based on $CIOU$. Through experimental testing, the value of the parameter α is most suitable when set to 3. When $\alpha = 3$, the $\alpha CIOU$ enables the model to focus more on high IOU targets, thereby improving the model's localization and detection performance. The improved loss function is shown in Eq. (12).

$$L_{\alpha IOU} = \frac{1 - IOU^\alpha}{\alpha} \quad (9)$$

$$L_{\alpha IOU} = \begin{cases} -\log(IOU), & \alpha \rightarrow 0 \\ 1 - IOU^\alpha, & \alpha > 0 \end{cases} \quad (10)$$

$$L_{\alpha IOU} = 1 - IOU^\alpha + \rho^\alpha(b, b^{gt}) \quad (11)$$

$$L_{\alpha CIOU} = 1 - IOU^\alpha + \frac{\rho^{2\alpha}(b, b^{gt})}{c^{2\alpha}} + \beta v)^\alpha \quad (12)$$

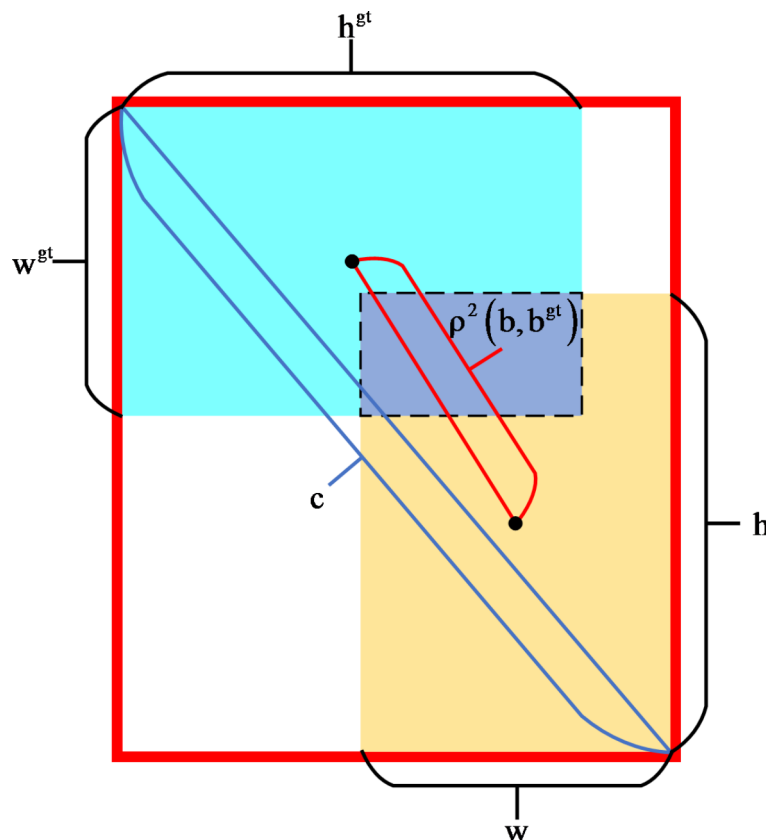


Fig. 7. CIOU loss calculation diagram.

Name	Configuration	Name	Configuration
CPU	12th Gen Intel(R)Core(TM) i7-12700KF 3.6GHZ	OS	Window 11
GPU	Nvidia GeForce RTX 3090Ti	Cuda version	CUDA 11.0
Memory	32 GB	Cudnn version	CUDNN 7.6.5
Graphics Memory	24 GB	Languge	Python 3.6

Table 1. Experimental environment configuration.

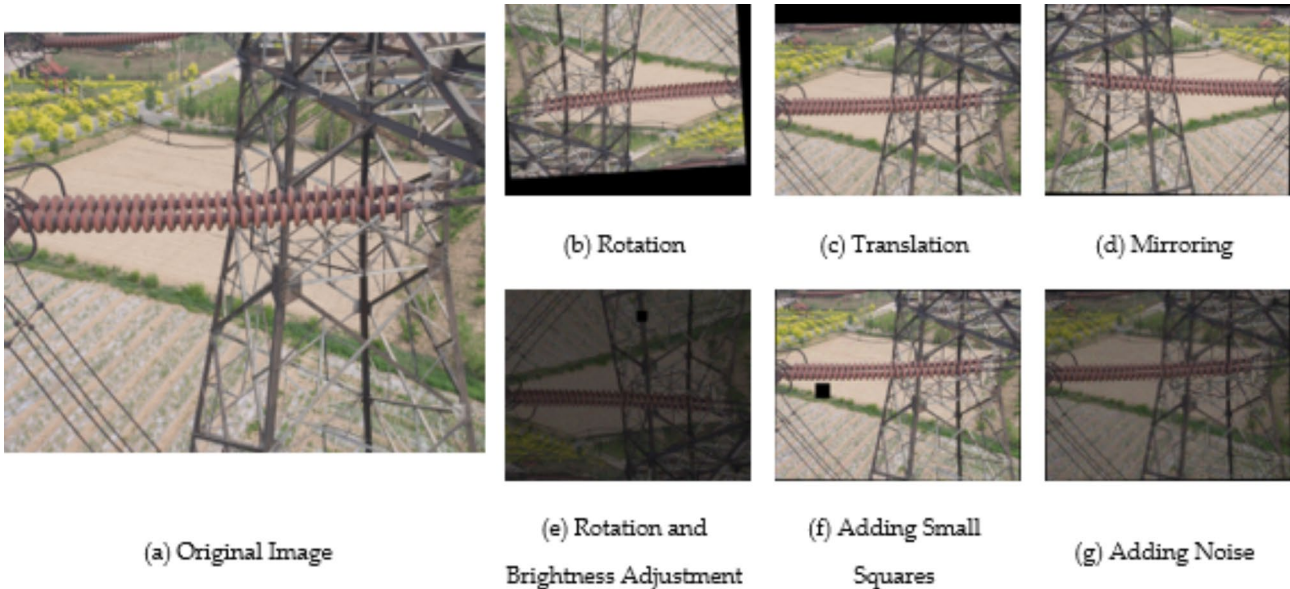


Fig. 8. Data preprocessing.

Experimental results and analysis
Experimental environment

The experiments in this paper are based on the PyTorch deep learning framework, using Python as the programming language for the experiment. All experiments were conducted on the Windows 11 operating system. The specific experimental environment details are provided in Table 1.

Data preprocessing

In this study, experimental data were gathered through aerial photography using drones, resulting in a total of 600 high-definition images of insulators. To enhance the diversity of the insulator samples and improve the model's generalization capability, a series of data augmentation techniques were applied to the original 600 images, including multi-directional rotation, translation, mirroring, brightness adjustment, addition of minor geometric elements, and noise injection, thereby expanding the dataset to 4,100 high-quality images. All images were uniformly resized to 640×640 pixels. The data preprocessing is shown in Fig. 8. The preprocessed dataset was manually annotated using the Labeling software and organized according to the VOC dataset format. To meet the experimental requirements, the dataset was split into training, validation, and test sets in a 7:2:1 ratio, with 2870 images in the training set, 820 images in the validation set, and 410 images in the test set. The training set was used for model training, the validation set was used to evaluate the model's performance, and the test set was mainly used for the final assessment of the model's inference speed.

Model training process

In this paper, the K-means++ clustering algorithm is used to generate 9 anchor boxes of 3 different scales. The sizes of the small-scale anchor boxes are (10, 13), (16, 30), and (33, 23); the medium-scale anchor boxes are (30, 61), (62, 45), and (59, 119); and the large-scale anchor boxes are (116, 90), (156, 198), and (373, 326). The DIOU-NMS non-maximum suppression algorithm is then used to filter the anchor boxes, removing redundant ones and retaining the anchor box with the maximum IOU value as the final detection box. The parameters of the improved YOLOv5 algorithm model are set as follows: the initial image size is set to 640×640, the number of training iterations is 200 epochs, the initial learning rate is 0.01, the batch size is set to 4, the momentum is set to 0.937, and the weight decay is set to 0.0005. The pseudocode for the model (Algorithm 1) is as follows:

Algorithm 1: Pseudocode for the modeInput: Original image of 640×640 pixels.

1. Using Mosaic data augmentation, randomly select four images to be stitched together into a new image of 640×640 pixels.
2. Employ the K-means++ clustering algorithm to generate suitable anchor boxes for this image size.
3. After setting the training parameters, input the processed image into the backbone network, neck network, and prediction network. The backbone and neck networks consist of a series of convolutional layers, normalization layers, and activation functions.
4. Specific operations in the prediction network include:
 - λ Classification scores: Generate a score for each predicted bounding box per category.
 - λ Bounding box regression: Generate the location information of bounding boxes, i.e., center coordinates and width and height.
 - λ Decode bounding boxes: Decode the model's output bounding box information into actual bounding box coordinates.
 - λ Non-Maximum Suppression (DIOU-NMS): Remove overlapping predictions and retain the most likely detection results.

Output: An image of 640×640 pixels containing the detection box results.

Figure 9 shows the loss curve plot of the improved YOLOv5 algorithm. Among these curves, the black and red lines represent the localization loss (box_loss) and confidence loss (obj_loss) during model training, respectively; the blue and green lines represent the localization loss and confidence loss during model validation, respectively. The localization loss (box_loss) indicates the error between the predicted bounding box and the ground truth bounding box (measured by $\alpha CIoU$); a smaller value indicates more accurate localization. The confidence loss (obj_loss) measures the network's accuracy in determining whether a target is present; a smaller value indicates stronger target classification ability. In Fig. 9, the x-axis represents the training iteration count. In this experiment, the model was trained for 200 iterations, and 200 training rounds were sufficient to achieve convergence. However, excessive iterations can lead to overfitting. The y-axis represents the loss value, which is the difference between the model's predicted output and the true labels. At the beginning of training, the loss value is typically high because the model has not yet learned how to fit the data. As training progresses, the loss

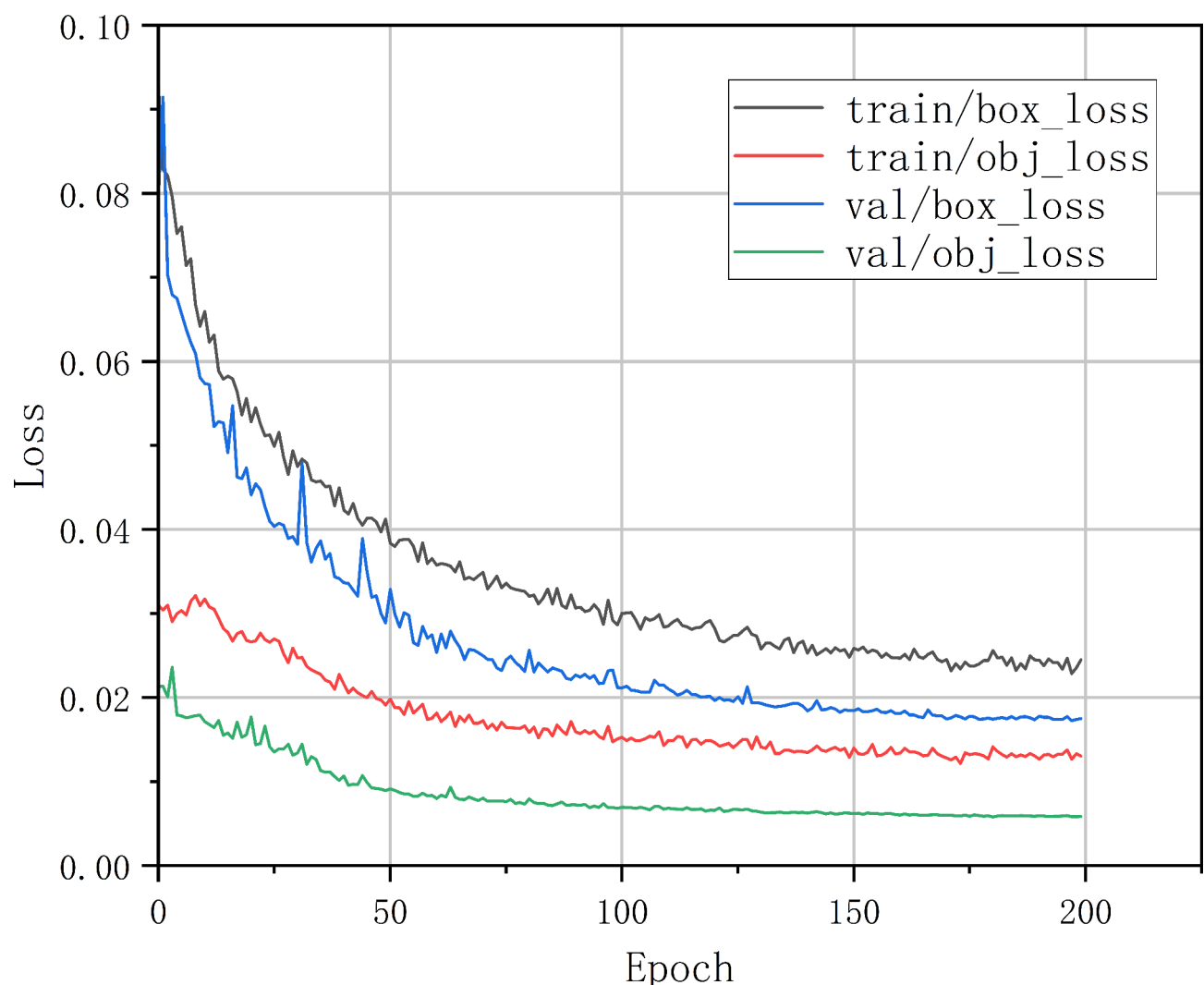


Fig. 9. The change curve of loss value after 200 iterations.

value gradually decreases, and the model's performance improves. When the loss value stabilizes and no longer shows significant decreases, it indicates that the model has converged.

Figure 10 shows the mAP curve after 200 training iterations. The x-axis represents the training iteration count, and the model was trained for 200 rounds, achieving convergence. The y-axis represents the average precision at different recall rates. mAP@0.5 is the average precision calculated at an IOU threshold of 0.5. The value 1.0 on the y-axis represents 100% recognition accuracy. A higher precision indicates that the model's predictions are more accurate, meaning a higher proportion of the predicted targets are correctly detected.

Evaluation metrics

The experiment primarily selects Precision (P), Recall (R), Average Precision (AP), mAP, parameter count, Frames Per Second (FPS) and Floating-point Operations Per Second (FLOPs) as the evaluation metrics for the performance of the YOLOv5 network model. P represents the proportion of truly positive samples among those identified by the model as positive. R represents the ratio of correctly identified positive samples to the total number of actual positive samples; a higher recall (R) indicates that more positive samples are correctly predicted by the model, suggesting better model performance. The mAP@0.5 denotes the mean average precision calculated when the IOU threshold is set to 0.5; a higher mAP@0.5 indicates better detection accuracy across all categories. The parameter count reflects the size of the model, and FLOPs are used to evaluate the computational complexity of the model; a larger parameter count and higher FLOPs indicate a more complex model with greater computational requirements. The FPS is primarily used to evaluate the real-time performance of a model. A high FPS indicates that the model can process more images within a unit of time, meaning that the model's inference speed is very fast. The specific calculation formulas are as follows:

$$P = \frac{TP}{TP + FP} \quad (13)$$

$$R = \frac{TP}{TP + FN} \quad (14)$$

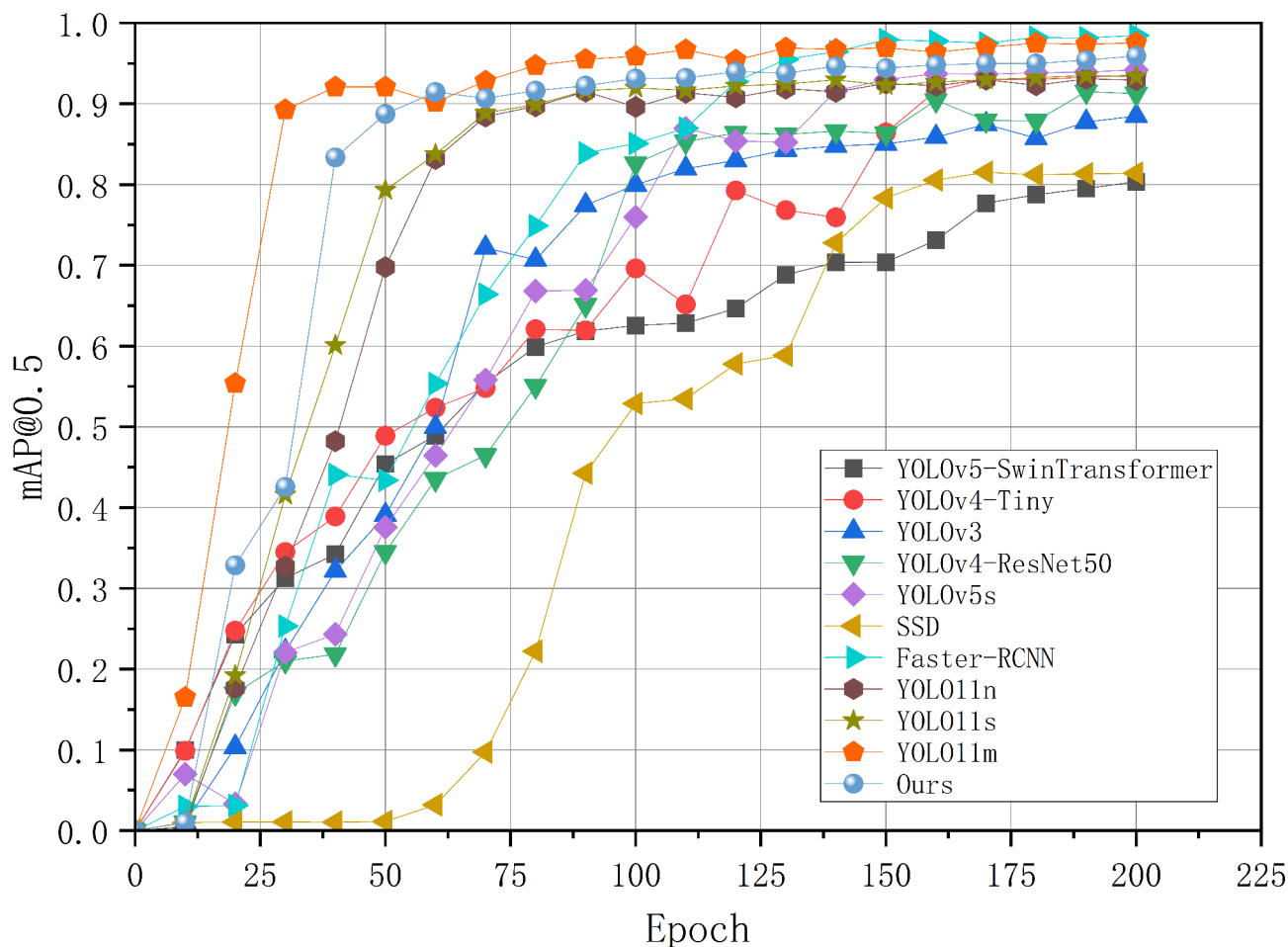


Fig. 10. The mAP value change curve after 200 iterations.

$$AP = \int_0^1 P(R) dR \tag{15}$$

$$mAP = \frac{\sum_{i=1}^N AP_i}{N} \tag{16}$$

Among these, TP (True Positives) represents the number of positive samples correctly identified by the model; TN (True Negatives) measures the number of instances correctly predicted as negative samples by the model; conversely, FP (False Positives) indicates the number of errors where the model incorrectly classifies negative samples as positive samples; FN (False Negatives) indicates the situation where positive samples are improperly classified as negative samples. N represents the number of categories of the samples; AP (Average Precision) reflects the identification accuracy of the model across different categories by calculating the integral of precision at different recall levels; and mAP, as the average of AP, summarizes the model's average precision performance across all categories, providing a comprehensive and intuitive measure of the overall effectiveness of the model^{28,29}.

Ablation experiment

To verify the impact of the improved method proposed in this paper on the performance of insulator object detection, five ablation experiments were set up for comparative analysis. Based on the YOLOv5 network model, the CSP-Darknet53 backbone network module and the lightweight CSP-SCConv backbone network module were compared. Then, the SPPF module, the RFB dilated convolution module, and the LSKBlock attention mechanism module were added one by one, using different combinations for training and testing. The test results are shown in Table 2.

As shown in Table 2, the improved YOLOv5 algorithm presented in this paper has improved in terms of parameter count, computational cost, and detection accuracy. Model 1 builds upon the original YOLOv5 by incorporating the LSKBlock attention mechanism, resulting in a reduction of 28.73% in the number of parameters and 42.37% in computational requirements. Compared to the original YOLOv5 model, this modification led to an accuracy improvement of 0.82%. The introduction of the LSKBlock attention mechanism not only effectively reduced the computational cost but also enhanced the model's accuracy. In comparison to the original YOLOv5 algorithm, Model 2 integrates both the RFB channel convolution module and the LSKBlock attention mechanism, achieving a decrease of 8.86% in the number of parameters and 11.34% in computational requirements. This indicates that the addition of these two modules can accelerate the training process. Compared to Model 1, Model 2 shows an accuracy improvement of 1.47%, demonstrating that the incorporation of the RFB channel convolution module further enhances the detection accuracy of the model. Model 3 replaces the backbone network CSP-Darknet53 with the lightweight CSP-SCConv module, reducing the parameter count and computational cost by 32.26% and 43.5%, respectively, indicating that the introduction of this module significantly reduces the parameter count and computational cost. Model 4, based on Model 3, discards the SPPF module and introduces the RFB channel convolution module and the LSKBlock attention mechanism module. It can be seen that the parameter count is reduced by 60.99% and the FLOPs is reduced by 74.03%, while the mAP improves by 3.77%. This indicates that the introduction of these three modules simultaneously reduces the parameter count and computational cost and enhances the detection accuracy of the model.

Experiment comparison

To further verify the performance of the proposed model, comparative experiments were conducted with single-stage object detection algorithms such as YOLOv3, YOLOv4-Tiny, YOLOv4-ResNet50, YOLOv5, YOLOv5-SwinTransformer, YOLO11n, YOLO11s and YOLO11m, as well as two-stage object detection algorithms like SSD and Faster-RCNN. The results are shown in Table 3.

As shown in Table 3, the improved YOLOv5-based insulator target detection algorithm proposed in this paper has achieved significant results in detection accuracy, reaching 95.60%, second only to Fast-er-RCNN and YOLO11m. Compared with other single-stage object detection algorithms such as YOLOv3, YOLOv4-Tiny, YOLOv4-ResNet50, YOLOv5, YOLOv5-SwinTransformer, YOLO11n, and YOLO11s, the proposed algorithm improves mAP@0.5 by 4.77%, 4.98%, 9.28%, 3.77%, 24.61%, 2.60%, and 2.00%, respectively, demonstrating a clear performance advantage. Additionally, compared to two-stage detection algorithms like SSD, the proposed algorithm improves mAP@0.5 by 19.3%, further highlighting its superiority in detection accuracy. In terms of parameter size and computational complexity, the proposed algorithm features a parameter size of 18.36 M and

Model	CSP-Darknet53	SPPF	CSP-SCConv	RFB	LSKBlock	Parameters (M)	FLOPs (G)	mAP@0.5 (%)
YOLOv5	√	√	×	×	×	47.06	115.92	91.83
Model 1	√	√	×	×	√	33.54	66.81	92.65
Model 2	√	×	×	√	√	42.89	102.77	94.12
Model 3	×	√	√	×	×	31.88	65.49	87.04
Model 4	×	×	√	√	√	18.36	30.10	95.60

Table 2. Ablation experiments based on improved YOLOv5.

Model	Backbone network	Parameters(M)	FLOPs (G)	P(%)	R(%)	FPS	mAP@0.5 (%)
YOLOv3	Darknet53	61.95	66.17	96.43	72.32	60.75	90.83
YOLOv4-Tiny	Darknet19	5.87	6.83	93.10	81.00	156.39	90.62
YOLOv4-ResNet50	ResNet50	33.25	35.05	88.89	72.00	73.29	86.32
YOLOv5	CSP-Darknet53	47.05	115.92	93.67	74.00	69.20	91.83
YOLOv5-SwinTransformer	Swin-Transformer	48.51	114.36	84.38	23.48	64.31	70.99
SSD	VGG16	23.61	60.76	93.50	58.08	111.53	76.30
Faster-RCNN	VGG16	136.68	401.69	64.10	99.98	28.56	98.60
YOLO11n	C3K2	2.50	6.40	93.70	86.90	162.88	93.00
YOLO11s	C3K2	9.40	21.50	91.60	90.8	156.25	93.60
YOLO11m	C3K2	20.03	67.60	92.80	94.2	142.85	97.60
Ours	CSP-SCConv	18.36	30.10	88.10	95.20	147.05	95.60

Table 3. Experimental results of different algorithms.

Model	Parameters(M)	FPS
YOLOv3	61.95	14.75
YOLOv4-Tiny	5.87	49.10
YOLOv4-ResNet50	33.25	24.91
YOLOv5	47.05	18.23
YOLOv5-SwinTransformer	48.51	16.92
SSD	23.61	38.16
Faster-RCNN	136.68	6.88
YOLO11n	2.5	52.07
YOLO11s	9.40	47.61
YOLO11m	20.03	41.13
Ours	18.36	43.24

Table 4. The FPS detection results of various algorithms on low-computing platforms.

a computational cost of 30.10G, which are only slightly higher than those of YOLOv4-Tiny, YOLO11n, and YOLO11s. However, when compared to other models, the parameter size and computational cost are significantly reduced, making the proposed algorithm more suitable for deployment on edge devices while maintaining high accuracy. The precision of the proposed algorithm reaches 88.10%, outperforming YOLOv5-SwinTransformer and Faster-RCNN. Although it may be slightly lower than some other models, the overall detection accuracy remains superior. Moreover, the recall rate of the proposed algorithm is 95.20%, second only to Faster-RCNN but higher than all other algorithms. This high recall rate ensures that more true targets are detected, reducing missed detections. The inference speed of the proposed algorithm is 147.05 FPS, ranking behind only YOLOv4-Tiny, YOLO11n, and YOLO11s, yet surpassing other algorithms. A higher frame rate allows for the processing of more images, enhancing the frequency of target detection and improving real-time performance. This makes the proposed algorithm especially suitable for real-time detection tasks on mobile platforms such as drones.

To enable the model to be deployed on devices with limited computational power, such as drones and embedded edge devices, reducing the model's parameter count can significantly enhance its operational efficiency on low-computing platforms. This not only effectively reduces computational costs but also improves inference speed and applicability. To validate the model's performance on low-computing platforms, experiments were conducted on an NVIDIA Jetson Orin Nano (8 GB RAM) processor to test the inference speeds of various models. As shown in Table 4, the improved model achieved an inference speed of 43.24 FPS, second only to YOLOv4-Tiny, YOLO11n, and YOLO11s. For real-time detection on low-computing platforms, an inference speed of at least 10–30 FPS is typically required to ensure stability during the detection process, and the improved model fully meets this requirement. Therefore, in real-time object detection tasks for drones, reducing the model's parameters provides strong support for deployment on devices with limited computational resources.

This paper selected insulator images with complex backgrounds for performance comparison. Figure 11 (a) to 11(k) show the detection results of different algorithms, each containing three insulators. Figure 11(a) to 11(f) all exhibit missed detections, while Fig. 11(g), 11(h), 11(i), 11(j), and 11(k) accurately detected all three targets. Among them, Fig. 11(g) achieved higher detection accuracy but is unsuitable for deployment on edge mobile devices due to its large parameter size and high computational cost. The detection accuracy of Fig. 11(j) is slightly higher than that of Fig. 11(k), but in terms of the number of parameters, computational complexity, and FPS, it does not match the proposed algorithm in this paper as ideally. In contrast, the detection accuracies of Fig. 11(h) and Fig. 11(i) are both lower than that of Fig. 11(k), making them insufficient to fully meet the requirements for high-precision detection.

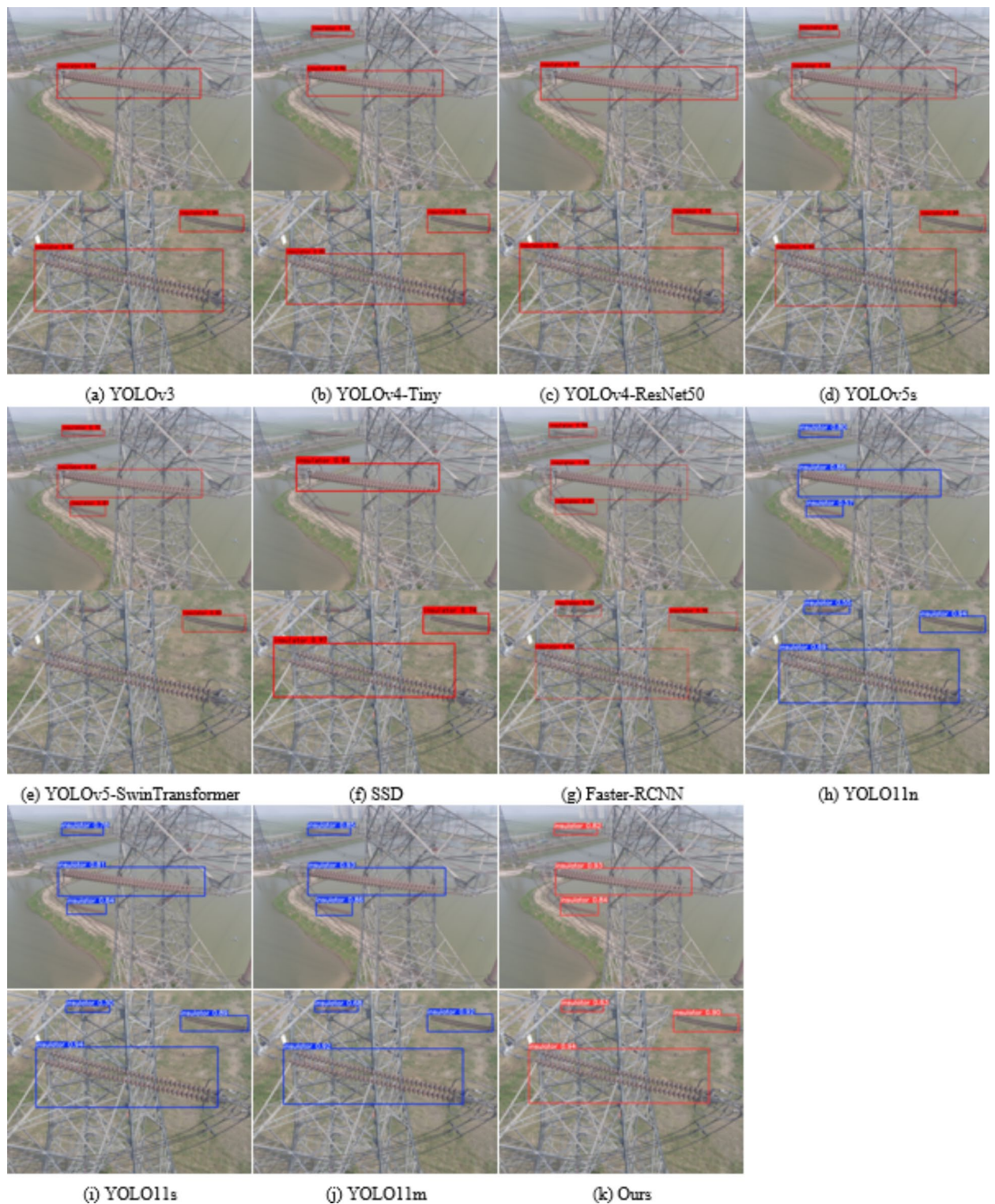


Fig. 11. Comparison of detection results of different algorithms. Figure 11(a) to 11(j) show the detection results of different algorithms. Figure 11(k) presents the detection results of the proposed algorithm in this paper.

Conclusions

To address the issues of low detection accuracy, high parameter count, and high miss-detection rates in traditional object detection algorithms, this paper improved the backbone network by incorporating ideas from the SCConv network, introduced the RFB channel convolution module and the LSKBlock attention mechanism module in the Neck network, and developed a lightweight model based on improved YOLOv5. The conclusions drawn from this study are as follows:

(1) By integrating the lightweight CSP-SCConv module into the backbone network, the feature extraction capability was significantly enhanced. This improvement not only enriched the feature information during the extraction process but also reduced redundant parameters.

(2) Utilizing the RFB channel convolution module and the LSKBlock attention mechanism within the feature fusion network enabled efficient integration of features from front to back, thereby improving the generalization ability of the network model.

(3) Incorporating the IOU-based loss function $\alpha CIOU$ dynamically adjusts the weights of the loss function and gradients based on the IoU value, thus improving the accuracy of bounding box regression and enhancing the robustness of the model.

Experimental results demonstrate that the improved YOLOv5 model achieves a detection accuracy of 95.60% for insulator targets, with reductions in parameter count and computational cost by 60.98% and 74.03%, respectively, compared to the original YOLOv5 model. This enhancement strongly supports real-time detection applications on edge devices used in drones. Although the YOLO11n and YOLO11s algorithms have fewer parameters and lower computational requirements, and their detection speeds are slightly higher than that of the improved YOLOv5 algorithm, the proposed improved algorithm outperforms YOLO11n and YOLO11s in terms of target detection accuracy and algorithm recall rate. In comparison, while the YOLO11m algorithm boasts higher detection accuracy, the proposed algorithm in this paper exhibits significant advantages in model parameter size, computational complexity, recall rate, and FPS. Insulators in transmission lines operate in complex environments, especially in scenarios involving complex mountainous regions and dense tree cover, where insulator targets are often subject to occlusion. In the future, the authors will conduct in-depth research focused on detecting insulator targets in these complex operating environments, aiming to further reduce the computational demands and parameter count of the algorithm while improving target detection accuracy. This not only enhances the application effectiveness of the algorithm model on edge intelligent terminals but also promotes its widespread adoption in practical deployments.

Data availability

The data generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

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Declarations

Competing interests

The authors declare no competing interests.

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